

Development Phase Roadmap

Iteration 1 -> Iteration 2 -> Iteration 3 | June 2026

Current State of Iteration 1 - What Exists

File	Purpose	State
file_ingestor.py	Reads PDF (PyMuPDF) and TXT/MD files, returns clean text.	DONE
gemini_pipeline_controller.py	Builds JSON-enforced Gemini prompt, calls google-genai SDK.	DONE
string_validator.py	Extracts JSON block, normalises dates/times, validates fields.	DONE
icalendar_factory.py	Builds RFC-5545 .ics file. Had a truncated class name -- FIXED in P0...	FIXED

All Phases - Ordered Delivery Plan

P0	Phase 0 - Bug Fix (COMPLETE) What: Repair truncated ExportResult class name + remove duplicate tail lines in icalendar_factory.py Files: src/export/icalendar_factory.py	DONE
P1A	Phase 1A - requirements.txt What: Add dependency file (pymupdf, google-genai, fastapi, uvicorn) so project is installable Files: requirements.txt (new file)	PENDING
P1B	Phase 1B - CLI Runner (main.py) What: Wire all 4 I1 modules into one command: python main.py syllabus.pdf Files: main.py (new file)	PENDING
P2	Phase 2 - FastAPI Backend What: POST /process: accept file upload, run full pipeline, return extracted events as JSON Files: server/main.py (new file)	PENDING
P3	Phase 3 - Frontend Upload Page What: Drag-and-drop HTML page with TailwindCSS CDN. Sends file to /process via fetch() Files: frontend/index.html (new file)	PENDING
P4	Phase 4 - Interactive Event Table What: Render editable rows per event. Fields: course, task, date, time, description + delete button Files: frontend/index.html (extend)	PENDING
P5	Phase 5 - ICS Download Trigger What: POST /download returns .ics stream. Browser Download button triggers file-save dialog Files: server/main.py + frontend/index.html (extend both)	PENDING
P6	Phase 6 - Integration Polish What: Test with a real PDF. Add graceful 429 rate-limit message and empty-syllabus fallback UI Files: server/main.py + frontend/index.html (extend both)	PENDING
P7	Phase 7 - Iteration 3 (Future) What: Google OAuth 2.0 direct push, rate-limit queue middleware, auto-retry error interceptor Files: server/ + frontend/ (extend)	PENDING

Iteration 2 - Component Detail (Phases 2 through 6)

Backend server/main.py -- Phases 2 and 5

- [POST /process]** Accepts multipart file upload. Reads file bytes into memory (never saves to disk -- privacy). Routes PDF bytes through PyMuPDF in-memory extractor -> Gemini pipeline -> JSON validator. Returns list of normalised event objects as JSON response.
- [POST /download]** Accepts the (possibly edited) events JSON sent from the browser table. Runs it through build_ics() from the existing icalendar_factory module. Returns the .ics string as an HTTP response with Content-Type: text/calendar and Content-Disposition: attachment so the browser saves it automatically.

Frontend frontend/index.html -- Phases 3, 4 and 5

- [Upload Zone]** Drag-and-drop file boundary styled with TailwindCSS CDN (no npm, no build step). Accepts PDF and TXT. Shows spinner/loading state while fetch() waits for /process.
- [Event Table]** Renders one editable row per extracted event. Columns: course name, task name, date (date picker input), time, description. Each row has a red delete button to remove incorrect entries.
- [Download Button]** Appears below the table after events load. POSTs the current table state (including any edits) to /download and the browser receives the .ics file as a download.
- [Error Banner]** Displays clean, user-readable messages for API errors, rate-limit 429 response, empty syllabus, or corrupted PDF. Never shows a raw stack trace to the user.

How to Continue This Project in a New Chat Session

- Step 1** Open a new Claude Code session in the same project folder.
- Step 2** Say: "Check memory and continue the Syllabus-to-Calendar project from where we left off."
- Step 3** Claude loads saved memory files from the .claude/projects/ folder and knows which phase is next.
- Step 4** To skip memory, just say the phase directly: "Start Phase 1" or "Start Phase 2" etc.
- Step 5** Memory folder: C:\Users\venkat\.claude\projects\...\memory\
- Step 6** Or paste this PDF into the new chat and say: "here is the blueprint, continue from Phase X"

TIP: The memory system remembers what is done, what is broken, and what is next. You never need to re-explain the project. Claude will pick up exactly where you left off.

Tech Stack Quick Reference

Library	Role
PyMuPDF (fitz)	PDF text extraction -- local, free, no external cloud needed
google-genai	Official Google Gen AI Python SDK for Gemini 1.5 Flash API calls
FastAPI + Uvicorn	Async Python web server for the /process and /download endpoints
TailwindCSS CDN	Utility CSS via <script> tag -- zero npm, zero build step
fpdf2	PDF generation for this plan document only -- not part of the app